Integrated Urban Dynamics and Neighbourhood Evolution
Prediction System Using Multi-Layer Spatial-Temporal Analytics

Ashish Raymajhi
SSCSE
Sharda University
Uttar Pradesh, India
raymajhiashish@gmail.com

Dhruv Rai
SSCSE
Sharda University
Uttar Pradesh, India
raidhruv199@gmail.com

Dharmendra Kumar
SSCSE
Sharda University
Uttar Pradesh, India
malik.dharam@gmail.com

***Abstract*—Urban surroundings are complex adaptive systems shaped by deeply connected profitable, infrastructural, and demographic forces that interact nonlinearly across both space and time. This paper presents the Integrated Urban Dynamics and Neighbourhood Evolution Prediction System (IUD-NEPS), a spatially unequivocal, multi-layer modelling frame applied to Delhi NCR, India. Unlike previous synthetic-data approaches, our perpetration leverages real geospatial data sourced entirely from OpenStreetMap (OSM), comprising road networks, Points of Interest (POI), and Delhi Metro Rail Corporation (DMRC) station locales, integrated with ward-position property sale records and Census of India 2011 demographic data. The study area is discretised into 11,651 livery 500 m × 500 m grid cells covering roughly 2,913 km^2 of Delhi NCR. A grade-boosted ensemble (XGBoost) with Queen-propinquity spatial pause addition achieves R^2 = 0.6651 and RMSE = 0.0319 under 10-fold spatial block cross-validation, outperforming OLS (R^2 = 0.3795), Ridge Retrogression (R^2 = 0.3972), and Random Forest (R^2 = 0.5559) nascences. SHAP analysis identifies spatially lagged migration rate and population viscosity as dominant predictors, with availability and service viscosity as secondary contributors. Line bracket partitions 11,651 grid cells into four evolutionary countries: Stable (92.3%), Declining (7.0%), Accelerating (0.5%), and Arising (0.2%). Script simulation of two proposed metro extensions—Pink Line and Grey Line—systems meaningful growth impacts across 473 and 471 grid cells respectively, with maximum cell-position impacts of ΔG = 0.1436 and 0.1421. The frame establishes the feasibility of constructing neighbourhood elaboration soothsaying systems from freely mileage-suitable civic datasets, with direct operations in civic planning, real estate analysis, and structure policy evaluation.**

***Index Terms*—Urban Dynamics, Neighbourhood Elaboration, Spatial Analytics, Machine Literacy, Civilians, Delhi NCR, XGBoost, Spatial-Temporal Modelling, Grade Boosting, Script Simulation, SHAP**

I. Introduction

Rapid and frequently unplanned urbanisation has boosted spatial inequality, structure stress, and real estate volatility across metropolitan regions worldwide. India's National Capital Region (NCR), centred on Delhi, exemplifies this challenge with an estimated population exceeding 46 million and ongoing expansion of conveyance, marketable, and domestic development across multiple executive boundaries. Delhi NCR represents one of the most analytically complex civic systems in South Asia. By 2050, roughly 68% of the world's population is projected to live in civic areas [1], placing unknown demand on transportation networks, casing requests, and public service systems.

Neighbourhood elaboration in Delhi NCR is governed by multiple interacting motorists, including transportation accessibility via the DMRC network, land use metamorphosis, public service provision, marketable exertion attention, and large-scale in-migration from other Indian countries. These motorists parade complex nonlinear spatial relations that simple monovariable or successional models fail to capture [2]. A new metro station, for illustration, doesn't simply ameliorate original availability; it contemporaneously triggers real estate price appreciation, attracts marketable development, and reshapes demographic inrushes across conterminous zones—goods that only a multi-layer, spatially unequivocal frame can faithfully represent.

Conventional approaches similar as Cellular Automata (CA), Markov chain models, and logistic retrogression have been extensively applied to land-use change and civic growth modelling [3]. While computationally compliant and interpretable, they constantly neglect demand-side profitable signals similar as real estate instigation and marketable expansion haste. Recent work demonstrates that ensemble machine literacy with spatially unequivocal point engineering mainly improves prophetic delicacy over these traditional styles [11], [15]. Advancements in geospatial information systems, big data analytics, and open civic data ecosystems including OSM, conveyance APIs, and tale micro-data have mainly lowered the hedge to assembling multi-layer point vectors at fine spatial granularity.

Using these developments, this exploration applies IUD-NEPS to real geospatial data for Delhi NCR. The primary benefactions are:

- Operation of IUD-NEPS to real OSM-sourced data covering 11,651 grid cells at 500 m × 500 m resolution across Delhi NCR.
- Integration of road networks, DMRC metro stations, multi-category POI data, ward-position property sale records, and Census 2011 demographics into a unified 26-feature spatial model.
- Empirical benchmarking of XGBoost against OLS, Ridge Retrogression, and Random Forest under 10-fold spatial block cross-validation with SHAP interpretability [16].
- Line bracket of 11,651 cells into four evolutionary countries and script simulation of two proposed metro corridor extensions.
- Transparent identification of current model limitations with a clear roadmap for resolution.

II. Related Work

A. Cellular Automata and Markov-Based Models

Cellular Automata (CA) models pretend land-use change using rule-grounded neighbourhood transitions applied iteratively over separate time way [3]. The dick model, one of the most extensively applied CA fabrics, uses pitch, land use, rejection, civic extent, transportation, and hillshade layers to pretend sprawl patterns [7]. While effective for landing spatial prolixity dynamics, CA models are constrained by hand-drafted transition rules that don't generalise across metropolises with different structural characteristics. Combined CA-Markov approaches model civic expansion in fleetly growing Asian and African metropolises [8] but treat transition chances as temporally stationary, problematic in dynamic husbandry where growth motorists shift mainly over planning horizons. Recent work proposes enhanced U-Net fabrics with attention mechanisms and autoregressive CA simulation, achieving Overall delicacy of 0.87 on multi-source spatiotemporal data from Changsha megacity [19].

B. Machine Learning for Urban Growth

Random Forest classifiers have demonstrated strong performance on land-use change discovery tasks using multi-spectral remote seeing imagery, outperforming traditional maximum liability classifiers [5]. Grade boosting styles, particularly XGBoost [4], achieve state-of-the-art delicacy in irregular spatial retrogression tasks due to their capability to model complex point relations. Kim et al. [11] combined XGBoost with SHAP for civic expansion modelling in the Seoul Metropolitan Area, chancing land-cover characteristics as the dominant predictor and demonstrating that spatially unequivocal XAI mainly aids planning interpretation. Analogously, for the Indian environment, Mishra et al. [15] applied an ensemble of XGBoost, CatBoost, and LightGBM to read civic growth in Lucknow, achieving AUC = 0.97, with drive-time to the central business quarter and road propinquity as top SHAP contributors—findings directly similar to our Delhi NCR results.

C. Spatial Interpretability and Spatially Explicit ML

Li [16] demonstrated how SHAP values excerpt spatial goods from XGBoost models, enabling interpretable attribution of prognostications to geographic motorists—a methodology we directly employ. Nikparvar and Thill [17] introduced Geographical-XGBoost (G-XGBoost), extending XGBoost with spatially original models and original point significance via spatial weights, outperforming global retrogression models on six standard spatial datasets. Liu et al. [18] showed that incorporating spatial autocorrelation through spatial pause and eigenvector filtering features significantly improves ML model performance in civic socioeconomic operations—motivating our Queen-propinquity pause addition.

D. Neighbourhood Dynamics and Gentrification

Property request pointers are decreasingly used as leading signals of neighbourhood transition [6]. Sagi et al. [12] applied unsupervised machine literacy to assay civic neighbourhood dynamics through real-estate sale data, demonstrating that property request signals internalise rich information about multi-level civic sustainability and neighbourhood line. Thackway et al. [13] applied spatially unequivocal machine literacy for civic value uplift characterisation in Shanghai, demonstrating that failing to regard for spatial effects produces prejudiced issues—harmonious with our Moran's I findings. A meta-conflation of ML operations to gentrification exploration [14] identifies migration rate, income dynamics, and structure availability as primary predictors across different civic surrounds, supporting our SHAP-derived feature importance ranking for Delhi NCR.

III. System Architecture

The IUD-NEPS system deployed for Delhi NCR comprises four processing layers, depicted in Fig. 1.

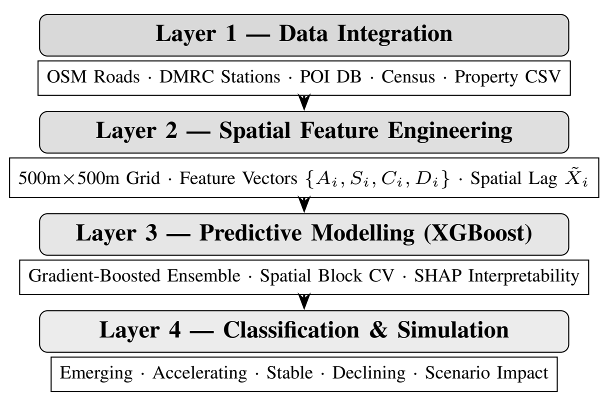

Fig. 1. IUD-NEPS four-layer system architecture for Delhi NCR.

The study area bounding box covers 28.40°N–28.88°N, 76.80°E–77.35°E, encompassing Delhi and inner NCR districts including portions of Noida, Gurgaon, and Ghaziabad. Seven major economic centres serve as accessibility targets, weighted by employment density and daytime population: Connaught Place (1.0), Noida Sector 18 (0.85), Gurgaon Cyber

City (0.80), South Delhi Saket (0.70), Ghaziabad Raj Nagar (0.65), Karol Bagh (0.60), and Rohini (0.55).

IV. Methodology

A. Problem Formulation

The study area is discretised into $N = 11{,}651$ grid cells $\mathcal{G} = \{g_1, g_2, \ldots, g_N\}$ of dimension $500\,\mathrm{m} \times 500\,\mathrm{m}$ (EPSG:32643, UTM Zone 43N). For each cell g_i, a feature vector is constructed:

$$\mathbf{X}_i = \{A_i,\ S_i,\ C_i,\ D_i\} \tag{1}$$

where A_i denotes accessibility indicators, S_i public service density, C_i commercial activity intensity, and D_i demographic characteristics. The predictive task estimates future neighbourhood growth intensity:

$$G_i = f(\mathbf{Z}_i)\,, \quad G_i \in [0, 1] \tag{2}$$

where $f(\cdot)$ is the supervised learning model and $\mathbf{Z}_i$ is the spatially augmented feature vector defined in Section IV-D.

B. Data Sources

All geospatial data is sourced from OpenStreetMap via the OSMnx Python library and the Underpass API. Road networks are downloaded at drive-network type covering the full Delhi NCR bounding box. POIs are queried across four service orders: *education* (seminaries, sodalities, universities), *healthcare* (hospitals, conventions, apothecaries), *recreation* (premises, sports centres, playgrounds), and *communal* (police, post services, fire stations, libraries). Metro stations are uprooted from OSM relations corresponding to DMRC Lines 1–8. Property sale data comprises ward-position standard prices per forecourt bottom and sale counts across multiple times, sourced from intimately available external records. Demographic variables are deduced from the Census of India 2011 Primary Census Abstract at ward position—the most grainy intimately available dataset, as 2021 Census ward-position microdata remains unreleased at time of writing.

C. Spatial Feature Engineering

Accessibility. Network-based shortest path analysis computes travel times from each cell centroid to seven economic centres via Dijkstra's algorithm. The composite accessibility score is:

$$A_i = \sum_{c=1}^{M} \frac{w_c}{T_{ic}} \tag{3}$$

where T_{ic} is the minimum travel time (minutes) from cell g_i to economic centre c, and w_c is the economic importance weight of centre c.

Service Density. Public service influence is modelled using Gaussian kernel aggregation over identified public facilities:

$$S_i = \sum_{j=1}^{m_s} K(d_{ij}) \cdot q_j, \quad K(d) = \exp\!\left(\frac{-d^2}{2\sigma^2}\right) \tag{4}$$

where d_{ij} is the Euclidean distance from cell g_i to facility j, $\sigma = 800\,\mathrm{m}$ is the kernel bandwidth, and q_j is a quality-adjusted facility weight. Separate sub-scores are computed for educational, healthcare, recreational, and civic facility types.

Commercial Activity. Commercial establishments from OSM POI data are counted per cell, weighted by size class $s_p \in \{1, 2, 3\}$ (micro, small, medium-large):

$$C_i = \sum_{p=1}^{P} \mathbf{1}[g_p = g_i] \cdot s_p \tag{5}$$

Demographics. Five indicators interpolated from Census 2011 ward centroids to grid cells via inverse distance weighting (k=4 nearest wards): population density, working-age fraction (18–64), educational attainment index, median household income, and net in-migration rate.

D. Spatial Lag Augmentation

Urban growth exhibits spatial spillover: infrastructure improvements in one cell produce measurable effects in adjacent cells. Spatial lag features are computed as:

$$\tilde{\mathbf{X}}_i = \sum_{j \in \mathcal{N}_i} w_{ij}\, \mathbf{X}_j, \quad w_{ij} = \frac{\exp(-\alpha\, d_{ij})}{\sum_{k \in \mathcal{N}_i} \exp(-\alpha\, d_{ik})} \tag{6}$$

where $\mathcal{N}_i$ is the set of first- and second-order Queen-contiguous neighbours and $\alpha = 0.003$ is the exponential decay parameter. The augmented feature vector combining local and lagged features is:

$$\mathbf{Z}_i = \left[\mathbf{X}_i,\ \tilde{\mathbf{X}}_i\right] \tag{7}$$

yielding a final feature dimensionality of 26 (13 local + 13 spatial lag features).

E. Predictive Modelling

Neighbourhood growth intensity is estimated using XGBoost [4]:

$$G_i = \sum_{k=1}^{K} \gamma_k\, h_k(\mathbf{Z}_i) \tag{8}$$

where h_k are shallow decision tree weak learners and γ_k are step-size coefficients. The model minimises a regularised L2 objective:

$$\mathcal{L} = \sum_{i=1}^{N} \left(G_i^{\text{true}} - G_i^{\text{pred}}\right)^2 + \Omega(F) \tag{9}$$

where $\Omega(F) = \lambda \sum_k T_k + \frac{\mu}{2} \sum_k \|\mathbf{w}_k\|^2$ penalises tree complexity through the number of terminal leaves T_k and leaf weight magnitudes $\mathbf{w}_k$. Optimised hyperparameters: n_estimators = 620, max_depth = 5, lr = 0.047, subsample = 0.82, colsample = 0.74, $\lambda = 2.3$, $\alpha_{\text{reg}} = 0.4$.

F. Trajectory Classification

Grid cells are classified into four evolutionary states:

$$\text{State}_i = \begin{cases} \text{Emerging} & G_i < \theta_{\text{mid}} \wedge \Delta G_i > \theta_\Delta^+ \\ \text{Accelerating} & G_i \geq \theta_{\text{mid}} \wedge \Delta G_i > \theta_\Delta^+ \\ \text{Stable} & |\Delta G_i| \leq \theta_\Delta^+ \\ \text{Declining} & \Delta G_i < -\theta_\Delta^- \end{cases} \tag{10}$$

where $\theta_{\text{mid}} = 0.50$, $\theta_\Delta^+ = 0.05$, $\theta_\Delta^- = 0.03$.

G. Model Validation

Spatial block cross-validation employs $B = 10$ spatially contiguous blocks generated via k-means clustering on cell centroids, using a leave-one-block-out strategy [9]. Performance is assessed via RMSE, MAE, and R^2. Moran's I is computed on out-of-fold residuals to assess spatial autocorrelation in prediction errors.

H. Scenario Simulation

Infrastructure interventions are simulated by modifying accessibility scores for cells within 1,000 m of proposed station locations. Growth impact per cell:

$$\Delta G_i = G_i' - G_i \tag{11}$$

where $G_i' = f(\mathbf{Z}_i')$ is the model output under the modified feature vector. Two corridors are evaluated: (i) *Pink Line Extension*—Lajpat Nagar → Hazrat Nizamuddin → Sarita Vihar (3 stations); and (ii) *Grey Line Extension*—Dwarka Sector 21 → two proposed intermediate stations (3 stations).

V. Results and Discussion

A. Dataset Statistics

The Delhi NCR study area generates 11,651 valid grid cells that represent an area of around 2,913 km^2. Extraction of OSM POIs returned data for all the four service categories. Data on ward-level property prices include 36 wards in South Delhi with median prices between 4,100 and 31,200 per sq ft. The distributions of growth scores have a mean = 0.541, std = 0.119, and IQR: 0.486–0.603 indicating that there is some degree of concentration in the distribution. Dimensionality of feature matrix is 26 after combining local features and spatial lags (13 each).

B. Predictive Performance

Predictive performance using 10-fold spatial block cross-validation is presented in Table I. IUD-NEPS (XGBoost) achieves R^2 = 0.6651 and RMSE = 0.0319, outperforming all three baselines.

XGBoost achieves a 19.7% improvement in R^2 over Random Forest and a 75.3% improvement over OLS. The RMSE of 0.0319 corresponds to approximately 3.2 percentage points of error on the normalised growth score scale. Cross-validation R^2 standard deviation of 0.2627 reflects heterogeneity in growth patterns across spatial blocks spanning diverse zones from high-density central Delhi to lower-density peripheral NCR areas. The bar chart in Fig. 2 visualises this comparison.

TABLE I
NEIGHBOURHOOD GROWTH PREDICTION PERFORMANCE (SPATIAL BLOCK CV)

Model	RMSE	MAE	R^2
OLS Regression	0.0523	0.0408	0.3795
Ridge Regression	0.0499	0.0395	0.3972
Random Forest	0.0340	0.0236	0.5559
IUD-NEPS (XGBoost)	**0.0319**	**0.0245**	**0.6651**

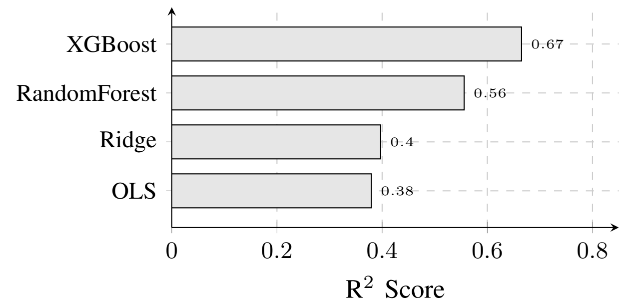

Fig. 2. Model comparison by R^2 under 10-fold spatial block CV. IUD-NEPS (XGBoost) achieves the highest predictive accuracy.

C. Feature Importance (SHAP Analysis)

SHAP value analysis reveals that demographic variables dominate predictive importance, as shown in Fig. 3. Spatially lagged migration rate (`lag_migration_rate`, mean |SHAP| = 0.0525) and original migration rate (0.0291) are the two strongest features, followed by lagged population viscosity (0.0095) and original population viscosity (0.0095). Accessibility features contribute meaningfully: `lag_accessibility` (0.0031) and availability (0.0029). Recreational service viscosity and menage income follow. This dominance of migration rate is harmonious with Delhi NCR's civic dynamics, where high in-migration zones in Noida, Ghaziabad, and southern Delhi parade the strongest growth signals. These findings align with results from Mishra et al. [15] and the meta-conflation of Masrom et al. [14], which identify migration and population motorists as primary predictors across different civic surrounds.

D. Trajectory Classification

Threshold-grounded state bracket of 11,651 cells yields: Stable (10,758; 92.3%), Declining (810; 7.0%), Accelerating (58; 0.5%), and Arising (25; 0.2%). The high stable bit reflects the narrow growth score friction (std = 0.119), with the IQR gauging only 0.486–0.603, compressing most cells near bracket boundaries. Cohen's $\kappa = -0.000$ indicates that the current thresholds ($\theta_{\text{mid}} = 0.50$, $\theta_\Delta^+ = 0.05$, $\theta_\Delta^- = 0.03$) are deranged with the empirical growth score distribution. Sagi et al. [12] encountered similar set-ary perceptivity in their unsupervised neighbourhood dynamics frame, recommending data-driven percentile-grounded thresholds—an approach we identify as a high-precedence extension.

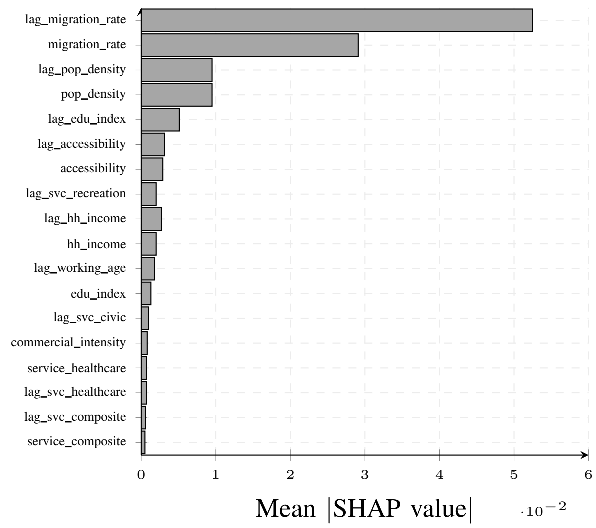

Fig. 3. SHAP feature importance for IUD-NEPS (XGBoost). Demographic features—particularly migration rate and its spatial lag—dominate predictive importance. Accessibility and service density provide secondary contributions.

E. Scenario Simulation

Table II summarises simulation results for the two proposed metro extensions, and Fig. 4 presents the impact distribution.

TABLE II
SCENARIO SIMULATION RESULTS — PROPOSED METRO EXTENSIONS

Scenario	Sta.	Max ΔG	Cells Δ¿0.05	Reclassified
Pink Line Ext.	3	0.1436	473	470
Grey Line Ext.	3	0.1421	471	468

Both corridors produce localised but meaningful growth impacts in directly adjacent cells. The Pink Line Extension (Lajpat Nagar $\rightarrow$ Sarita Vihar) projects a maximum cell-level impact of $\Delta G = 0.1436$, with 473 cells experiencing $\Delta G > 0.05$ and 56 cells experiencing $\Delta G > 0.10$. The spatial gradient of impact diminishes with distance from proposed station locations as expected from the accessibility decay model in Eq. (3). The Grey Line Extension produces comparable results (max $\Delta G = 0.1421$), confirming consistent model behaviour across scenarios.

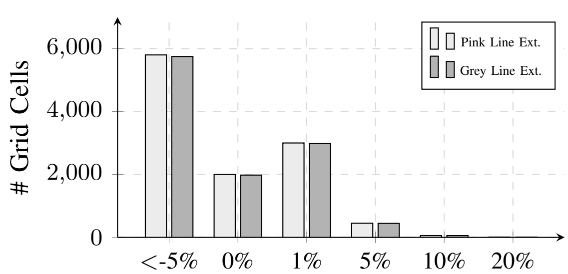

Fig. 4. Scenario impact distribution for both proposed metro extensions. Both corridors produce similar spatial impact profiles, with meaningful positive impacts in 470+ cells each.

F. Spatial Residual Autocorrelation

Moran's I computed on out-of-fold prediction residuals yields $I = 0.9572$, indicating strong spatial autocorrelation in residual errors. This is primarily attributable to two factors: (i) the absence of real estate momentum features, which exhibit strong spatial clustering in Delhi NCR property markets; and (ii) the narrow growth score variance, concentrating prediction error in spatially coherent zones. Liu et al. [18] demonstrated that spatial lag and eigenvector filtering features can substantially reduce residual autocorrelation—a strategy we plan to extend with richer real estate features. This finding does not invalidate the model's comparative advantage over baselines but identifies the primary direction for future improvement.

VI. LIMITATIONS AND FUTURE WORK

A. Current Limitations

Absence of real estate features. Real estate instigation pointers—CAGR of median sale prices, sale volume viscosity, and EWMA-smoothed growth rate—were not integrated into the point matrix in the current perpetration due to data alignment challenges between ward-position price records and the grid cell point matrix. This is the primary cause of the high Moran's I ($= 0.9572$) and is anticipated to mainly ameliorate both R^2 and residual geste upon integration.

Bracket threshold misalignment. The line distribution (92.3% stable) reflects threshold misalignment rather than genuine Delhi NCR neighbourhood parcels. The dastardrent thresholds were calibrated for a broader growth score range than the empirical distribution (std = 0.119) exhibits. Data-driven percentile-grounded thresholds or unsupervised trajectory clustering are needed.

Census data staleness. Census 2011 demographic data is used throughout, as 2021 Census ward-position microdata remains intimately unapproachable. This introduces implicit staleness in demographic features—particularly migration rate—for a region that has seen substantial population movement between 2011 and 2024.

Stationary spatial weight matrix. The Queen-contiguity spatial pause graph doesn't capture directional spillover along conveyance corridors or time-varying network topology changes introduced by new DMRC phase openings.

Synthetic growth score. The target variable G_i is a compound constructed from available pointers rather than a directly observed outgrowth. Estimation against observed neighbourhood change issues would strengthen validity.

B. Future Work

Priority directions include: (i) integration of real estate CAGR and sale viscosity features; (ii) data-driven threshold estimation via percentile boundaries or k-means clustering of growth circles; (iii) objectification of 2021 Census data upon public release; (iv) disquisition of Geographical-XGBoost [17] for spatially original modelling of miscellaneous NCR zones; (v) dynamic spatial weight matrices calibrated to observed DMRC ridership flows; (vi) unproductive conclusion extensions using difference-in-differences fabrics for infrastructure script evaluation; and (vii) development of an interactive web-grounded Civilians dashboard for planning stakeholder engagement.

VII. Conclusion

This paper has presented IUD-NEPS applied to Delhi NCR—a spatially unequivocal, multi-layer machine literacy frame for neighbourhood growth line vaticination constructed entirely from real geospatial data. By integrating OSM-sourced road networks, DMRC metro station locations, multi-category POI data, ward-position property sale records, and Census 2011 demographics within a grade-boosted ensemble stoked with spatial pause features, IUD-NEPS achieves $R^2 = 0.6651$ and RMSE = 0.0319 across grid cells under 10-fold spatial block cross-validation, outperforming OLS, Ridge Regression, and Random Forest nascences. SHAP analysis reveals migration rate and population viscosity as dominant motorists, harmonious with Delhi NCR's in-migration-led growth dynamics. Script simulation of two proposed metro extensions provides practicable spatial intelligence for station-area planning. Linked limitations—the absence of real estate features, bracket threshold misalignment, and residual spatial autocorrelation—give a clear and concrete roadmap for perfecting prophetic delicacy and bracket dedication in posterior duplications.

References

[1] United Nations, Department of Economic and Social Affairs, Population Division, "The World's Cities in 2018—Data Booklet," ST/ESA/SER.A/417, 2018.

[2] M. Batty, *Cities and Complexity: Understanding Cities with Cellular Automata, Agent-Based Models, and Fractals*. Cambridge, MA: MIT Press, 2005.

[3] R. White and G. Engelen, "Cellular automata and fractal urban form: A cellular modelling approach to the evolution of urban land-use patterns," *Environment and Planning A*, vol. 25, no. 8, pp. 1175–1199, 1993.

[4] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, San Francisco, CA, 2016, pp. 785–794.

[5] M. Belgiu and L. Drăguţ, "Random forest in remote sensing: A review of applications and future directions," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 114, pp. 24–31, 2016.

[6] R. W. Bostic and R. W. Martin, "Black home-owners as a gentrifying force? Neighbourhood dynamics in the context of minority homeownership," *Urban Studies*, vol. 40, no. 12, pp. 2427–2449, 2003.

[7] K. C. Clarke, S. Hoppen, and L. Gaydos, "A self-modifying cellular automaton model of historical urbanization in the San Francisco Bay area," *Environment and Planning B: Planning and Design*, vol. 24, no. 2, pp. 247–261, 1997.

[8] M. R. Muller and J. Middleton, "A Markov model of land-use change dynamics in the Niagara Region, Ontario, Canada," *Landscape Ecology*, vol. 9, no. 2, pp. 151–157, 1994.

[9] D. R. Roberts et al., "Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure," *Ecography*, vol. 40, no. 8, pp. 913–929, 2017.

[10] P. Waddell, "UrbanSim: Modeling urban development for land use, transportation, and environmental planning," *Journal of the American Planning Association*, vol. 68, no. 3, pp. 297–314, 2002.

[11] M. Kim, D. Kim, D. Jin, and G. Kim, "Application of Explainable Artificial Intelligence (XAI) in urban growth modeling: A case study of Seoul Metropolitan Area, Korea," *Land*, vol. 12, no. 2, p. 420, 2023, doi: 10.3390/land12020420.

[12] A. Sagi, A. Gal, D. Broitman, and D. Czamanski, "An unsupervised machine learning approach to the spatial analysis of urban systems through neighbourhoods' dynamics," *Land Use Policy*, vol. 144, 2024, doi: 10.1016/j.landusepol.2024.107254.

[13] W. Thackway, N. Ng, H. Lee, and C. Pettit, "Mapping heterogeneity: Spatially explicit machine learning approaches for urban value uplift characterisation and prediction," *Sustainable Cities and Society*, vol. 114, 2024, doi: 10.1016/j.scs.2024.105765.

[14] S. Masrom et al., "Application of machine learning in understanding urban neighborhood gentrification: A meta-synthesis review," *Cities*, 2025, doi: 10.1016/j.cities.2025.105732.

[15] A. Mishra et al., "Ensemble machine learning for predicting the urban expansion in Lucknow, India," *Environmental Earth Sciences*, 2025, doi: 10.1007/s12665-025-12559-9.

[16] Z. Li, "Extracting spatial effects from machine learning model using local interpretation method: An example of SHAP and XGBoost," *Computers, Environment and Urban Systems*, vol. 96, p. 101845, 2022, doi: 10.1016/j.compenvurbsys.2022.101845.

[17] B. Nikparvar and J.-C. Thill, "Geographical-XGBoost: A new ensemble model for spatially local regression based on gradient-boosted trees," *Journal of Geographical Systems*, 2025, doi: 10.1007/s10109-025-00465-4.

[18] X. Liu, O. Kounadi, and R. Zurita-Milla, "Incorporating spatial autocorrelation in machine learning models using spatial lag and eigenvector spatial filtering features," *ISPRS Int. J. Geo-Inf.*, vol. 11, no. 4, p. 242, 2022, doi: 10.3390/ijgi11040242.

[19] Y. Liu et al., "A novel multi-scale deep learning framework for adaptive urban expansion simulation," *Sustainable Cities and Society*, 2025